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Key Components of Llamafile

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Reading: Cosmopolitan
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Developing Portable CLI with Cosmopolitan

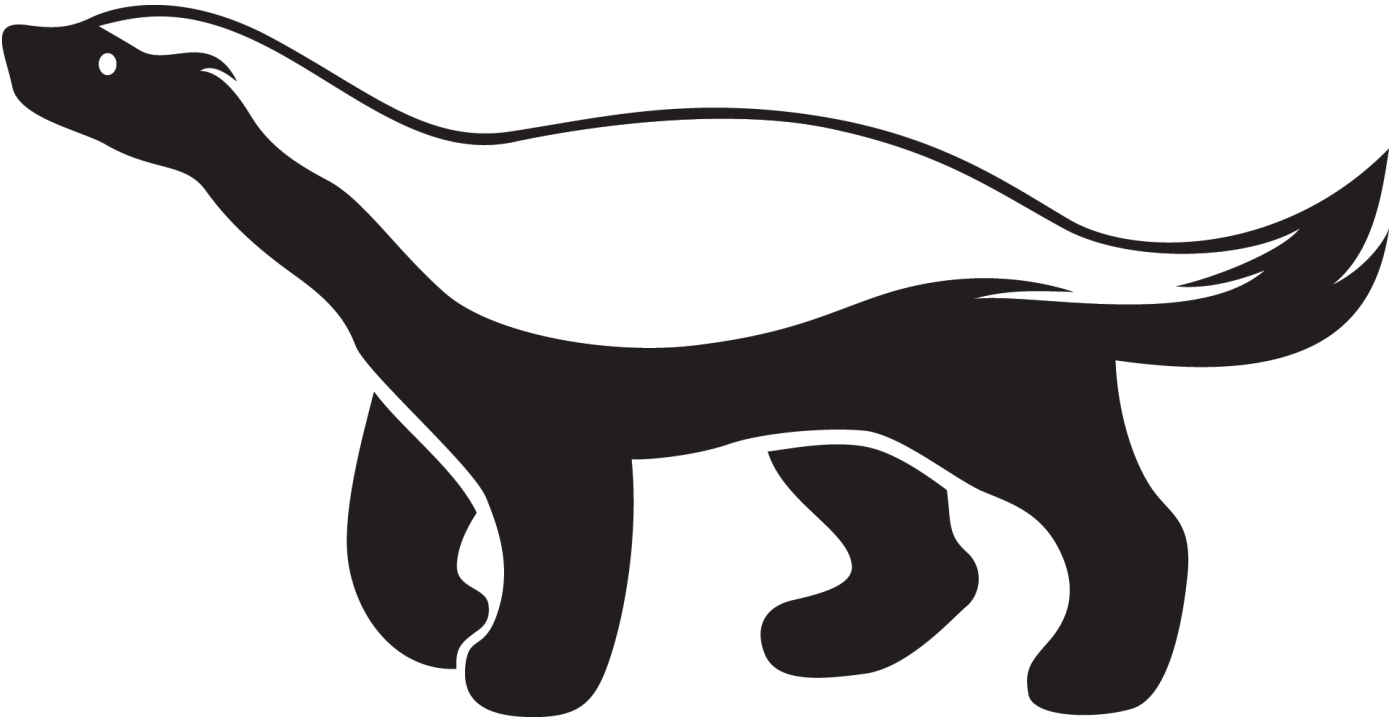
Running Llamafile

Conclusion and Next Steps

Cosmopolitan

Summary:

The [Cosmopolitan projec](#) [↗]t aims to make C a build-once, run-anywhere language like Java, but without needing an interpreter or virtual machine. It does this by reconfiguring GCC/Clang to output a polyglot format that runs natively on Linux, Mac, Windows and other operating systems. Key points are building and debugging Cosmopolitan programs, platform specific notes, and minimum OS version support.



Key Points:

- Use cosmocc compiler to build C programs that run anywhere
- Print system call traces for debugging with --strace
- Register binfmt_misc on Linux to run programs properly
- Disable WSLInterop on WSL for safety
- Works on old operating systems like Windows 8 and Linux 2.6.18

Reflection Questions:

1. What problem does the Cosmopolitan project try to solve?
2. How does it provide "build once, run anywhere" portability?
3. What debugging features does Cosmopolitan provide?
4. What potential issues are there running Cosmopolitan programs on Linux or WSL?
5. What is the oldest operating system version supported?

Challenges:

1. Try compiling and running a simple Cosmopolitan "Hello World" program.
2. Print a system call trace or function trace to see debugging info.
3. Build a more complex open source program using Cosmopolitan's toolchain.
4. Get Cosmopolitan programs working properly on WSL or an old Linux distro.
5. Browse the source code to understand how the polyglot format works.